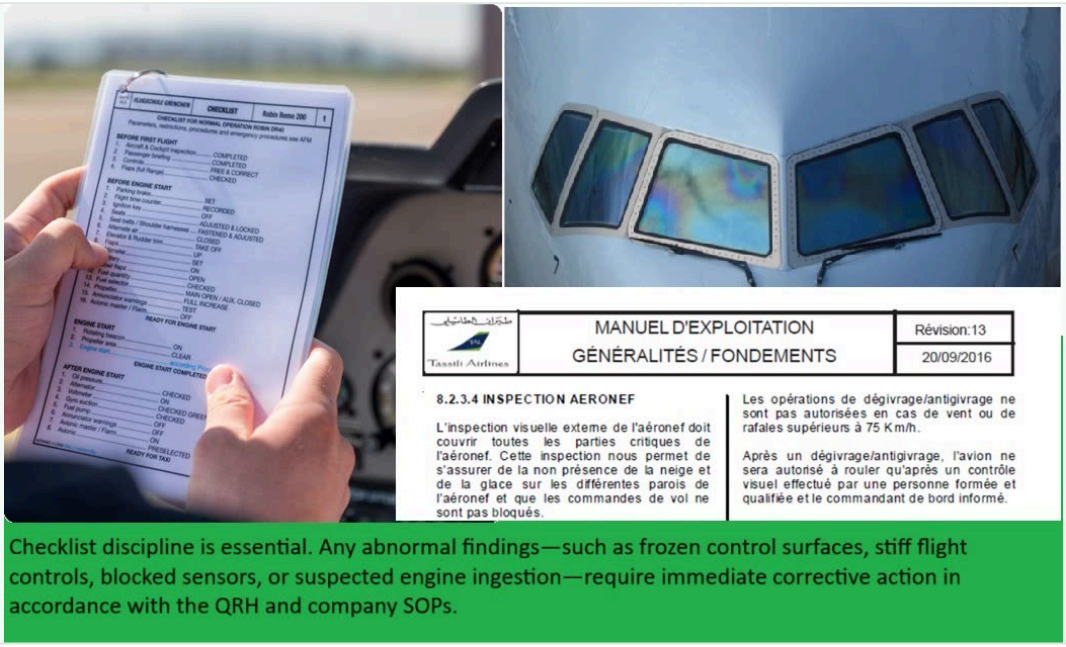
MODULE 5 · AIRCRAFT SYSTEMS IN THE COLD

Lesson 19: Cold Soak, Overnight Parking and Pre-flight Preparation

Lesson 19 of 30 · about 8 min

LEARNING OBJECTIVES

- Protect an aircraft parked overnight in cold conditions.
- Prepare a cold-soaked aircraft for flight.
- Identify the effects of cold on tyres, brakes, water systems and batteries.



Cold-weather preparation follows the checklist and the Operations Manual — and takes extra time.

Cold soak

An aircraft is **cold-soaked** when its structure, fluids and equipment have cooled to the ambient temperature — usually after several hours parked outside in the cold, or after a long flight at high altitude. Cold soak affects nearly every system: fluids become thicker, seals harden, batteries lose capacity, and water freezes.

Overnight parking (as the company procedure and AFM specify)

- **Covers and plugs** fitted on engines, probes and inlets where available. Record them so they are removed before flight.
- **Potable water and waste systems** drained if the aircraft will be unheated, to prevent pipes, tanks and valves from freezing.
- **Parking brake**: after taxiing through slush or snow, wet brakes can freeze. The common practice is to chock the wheels and release the parking brake when permitted.
- **Batteries**: in extreme cold, some types require removal or a heated location.

- **Fuel**: drain water from the tank sumps as maintenance requires.
- **Doors closed** to keep snow out and reduce cabin cooling.
- Where possible, park the aircraft nose into the wind or in a hangar when heavy snow or freezing precipitation is forecast.

Preparing a cold-soaked aircraft

- Plan for extra time. Cabin heating, de-icing and systems warm-up all take longer.
- Check that all covers and plugs are removed and that no snow or ice remains in inlets, wheel wells, vents and drains.
- Check tyre pressures. Pressure falls by about **1 % for every 3 °C** drop in temperature, so a tyre set in a warm hangar will be under-inflated outside.
- Hydraulic, oil and strut indications may read differently when cold. Follow the servicing procedures for cold temperatures.
- Oxygen bottle pressure is lower in the cold. Use the temperature-corrected minimum pressure chart.
- Switch on window heat early, at the low setting if your type has one, to avoid thermal shock to cold windshields.
- Warm the cabin before boarding when possible, using the APU or ground air conditioning.

KEY TAKEAWAYS

- Cold soak affects fluids, seals, batteries, water systems and tyres.
- Drain water systems, fit covers, and chock with the parking brake released after slush taxi (as the SOP states).
- Tyre pressure drops about 1 % per 3 °C — check it outside.
- Plan extra time for the preparation of a cold-soaked aircraft.

KNOWLEDGE CHECK

A tyre was set to the correct pressure at +15 °C in a hangar. The aircraft is now outside at −15 °C. What do you expect?

A. Pressure about 10 % lower

B. No change in pressure

C. Pressure about 10 % higher

D. Pressure about 30 % lower

Answer: A. A 30 °C drop at about 1 % per 3 °C gives roughly a 10 % reduction in tyre pressure.

Training material only. It does not replace the Aircraft Flight Manual, FCOM, QRH, MEL or the Domestic Airlines Operations Manual, which always take precedence.